



National  
Qualifications  
2015

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# **2015 Geography**

## **National 5**

### **Finalised Marking Instructions**

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## General Marking Principles for National 5 Geography

*This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this Paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.*

- (a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.
- (b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.
- (c) If a specific candidate response does not seem to be covered by either the principles or detailed Marking Instructions, and you are uncertain how to assess it, you must seek guidance from your Team Leader.
- (d)
  - (i) For credit to be given, points must relate to the question asked.
  - (ii) There are six types of question used in this question paper, namely:
    - A. Describe . . .
    - B. Explain . . .
    - C. Give reasons . . .
    - D. Match . . .
    - E. Give map evidence . . .
    - F. Give advantages and/or disadvantages . . .
  - (iii) For each of the question types in this paper, the following provides an overview of marking principles and an example of its application for each question type.

### Questions that ask candidates to **Describe** . . . (4-6 marks)

Candidates must make a number of relevant, factual points. These should be key points. The points do not need to be in any particular order. Candidates may provide a number of straightforward points or a smaller number of developed points, or a combination of these.

Up to the total mark allocation for this question:

- **One mark** should be given for each accurate relevant point.
- **Further marks** should be given for development and exemplification.

**Question:** Describe, in detail, the effects of two of the factors shown. (Modern factors affecting farming).

#### **Example:**

New technology has led to increased crop yields (**one mark**), leading to better profits for some farmers (**a second mark for development**).

### Questions that ask candidates to **Explain** . . . (4-6 marks)

Candidates must make a number of points that make the process/situation plain or clear, for example by showing connections between factors or causal relationships between events or processes. These should be key reasons and may include theoretical ideas. There is no need for any prioritising of these reasons. Candidates may provide a number of straightforward reasons or a smaller number of developed reasons, or a combination of these. The use of the command word 'explain' will generally be used when candidates are required to demonstrate knowledge

and understanding. However, depending on the context of the question the command words 'give reasons' may be substituted.

If candidates produce fully labelled diagrams they may be awarded up to full marks if the diagrams are sufficiently accurate and detailed.

Up to the total mark allocation for this question:

- **One mark** should be given for each accurate relevant point.
- **Further marks** should be given for developed explanations.

**Question:** Explain the formation of a U-shaped valley.

**Example:**

A glacier moves down a main valley which it erodes (**1 mark**) by plucking, where the ice freezes on to fragments of rock and pulls them away. (**second mark for development**).

#### **Questions that ask candidates to *Give reasons* . . . (4-6 marks)**

Candidates must make a number of points that make the process/situation plain or clear, for example by showing connections between factors or causal relationships between events or processes. These should be key reasons and may include theoretical ideas. There is no need for any prioritising of these reasons. Candidates may provide a number of straightforward reasons or a smaller number of developed reasons, or a combination of these. The use of the command words 'give reasons' will generally be used when candidates are required to use information from sources. However, depending on the context of the question the command word 'explain' may be substituted.

Up to the total mark allocation for this question:

- **One mark** should be given for each accurate relevant point.
- **Further marks** should be given for developed reasons.

**Question:** Give reasons for the differences in the weather conditions between Belfast and Stockholm.

**Example:**

In Stockholm it is dry but in Belfast it is wet because Stockholm is in a ridge of high pressure whereas Belfast is in a depression (**one mark**). Belfast is close to the warm front and therefore experiencing rain (**second mark for development**).

#### **Questions that ask candidates to *Match* (3-4 marks)**

Candidates must match two sets of variables by using their map interpretation skills.

Up to the total mark allocation for this question:

**One mark** should be given for each correct answer.

**Question:** Match the letters A to C with the correct features.

**Example:** A = Forestry (**1 mark**)

#### **Questions that ask candidates to *Give map evidence* (3-4 marks)**

Candidates must look for evidence on the map and make clear statements to support their answer.

Up to the total mark allocation for this question:

**Question:** Give map evidence to show that part of Coventry's CBD is located in grid square 3379.

**Example:** Many roads meet in this square (**1 mark**).

**Questions that ask candidates to *Give advantages and/or disadvantages* (4-6 marks)**

Candidates must select relevant advantages or disadvantages of a proposed development and show their understanding of their significance to the proposal. Answers may give briefly explained points or a smaller number of points which are developed to warrant further marks.

Up to the total mark allocation for this question:

- **One mark** should be given for each accurate relevant point.
- **Further marks** should be given for developed points.
- Marks should be awarded for accurate map evidence.

**Question:** Give either advantages or disadvantages of this location for a shopping centre. You must use map evidence to support your answer.

**Example:** There are roads and motorways close by allowing the easy delivery of goods(**1 mark**) and access for customers (**1 mark for development**), eg the A46, M6 and M69.

## Detailed Marking Instructions for each question

### Section 1: Physical Environments

| Question |     |  | General Marking Instructions for this type of question                                                                                                                                                                                          | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.       | (a) |  | 3 correct = 3 marks<br>2 correct = 2 marks<br>1 correct = 1 mark.                                                                                                                                                                               | 3        | Headland - 766356<br>cliff - 690382<br>bay - 674398                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|          | (b) |  | One mark for each valid point.<br>Two marks for a developed point.<br>Well-annotated diagrams can obtain full marks.<br>Answers will depend on choice.<br><b>Maximum 1 mark for a list of processes</b><br>Do not credit references to a stump. | 4        | <p><b>Stack</b><br/>Waves attack a line of weakness in the headland (1). Types of erosion include hydraulic action, corrosion and corrasion (1). Continuous erosion will open up the crack and it will develop into a sea cave (1). Further erosion of the cave, often on opposite sides of the headland, will form an arch (1). The roof of the arch is attacked by the waves until it eventually collapses (1). This leaves behind a free standing piece of rock called a stack which is separate from the headland (1).<br/>Or any other valid point.</p> <p><b>Bay</b><br/>Bays are formed due to differential erosion (1) where rocks along the coastline are formed in alternating bands of different rock types (1) eg sandstone and clay (1) and which meet the coast at right angles (1). Clay is a softer rock than sandstone so it is eroded more quickly (1). The waves erode the softer rock through hydraulic action, corrasion and corrosion (1) to form sheltered bays (1) which may have beaches (1). The harder sandstone areas are more resistant to erosion and jut out into the sea to form exposed headlands (1).<br/>Or any other valid point.</p> |

| Question |     |  | General Marking Instructions for this type of question                                                                                                                                                                                                            | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| 2.       | (a) |  | 3 correct = 3 marks<br>2 correct = 2 marks<br>1 correct = 1 mark.                                                                                                                                                                                                 | 3        | Levéé - 684466<br>Meander - 708473<br>V-Shaped Valley - 713410                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|          | (b) |  | One mark for each valid point.<br>Two marks for a developed point.<br>Well-annotated diagrams can obtain full marks.<br>Answers will depend on choice.<br><b>One mark for reference to a list of processes of erosion such as corrosion and hydraulic action.</b> | 4        | <p><b>Meander</b><br/>In the middle/lower course, a river flows downhill causing lateral erosion (1). The river contains areas of deep water and areas of shallow water, this results in areas of slower and faster water movement and this causes the current to swing from side to side (2). The river flows faster on the outer bank and erodes it (1). This forms a river cliff (1). The river flows more slowly on the inner bank and deposits some of its load (1). This forms a river beach/slip-off slope (1). Continuous erosion on the outer bank and deposition on the inner bank forms a meander in the river (1).<br/>Or any other valid point.</p> <p><b>V-shaped valley</b><br/>In the upper course, a river flows downhill eroding the landscape vertically (1). The river erodes a deep notch into the landscape using hydraulic action, corrasion and corrosion (1). As the river erodes downwards the sides of the valley are exposed to freeze-thaw weathering which loosens the rocks and steepens the valley sides (2). The rocks which have fallen into the river aid the process of corrasion which leads to further erosion (1). The river transports the rocks downstream and the channel becomes wider and deeper creating a V-shaped valley between interlocking spurs (2).<br/>Or any other valid point.</p> |

| Question |  |  | General Marking Instructions for this type of question                                                                                                                                                                                                                                                                                                                                                            | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 3.       |  |  | <p>One mark for each valid point.<br/>Two marks for a developed point.<br/>Maximum one mark for a correct grid reference/named example.</p> <p><b>No marks for description.</b> Answers must be explanations.</p> <p>Maximum of 3 marks if only one land use is mentioned. Mark as 3:2 or 2:3.</p> <p>If suitability of more than two land uses is explained, then award marks for the best two explanations.</p> | 5        | <p>Answers will vary depending upon the land uses chosen.</p> <p><b>For farming:</b> Reads Farm(1) (at grid reference 728489) is an example of a hill sheep farm as the land is steep (1). As the land is higher up, the weather will be harsh and sheep can survive these conditions, especially in winter (1). The land is too steep for farm machinery to operate (1). The soil will be too thin for crops to be grown (1).</p> <p><b>For tourism and recreation:</b> The South West Coastal Path follows the top of the cliffs and allows tourists to enjoy a view of the coastal scenery (1) eg 727367 (1). There is a nature reserve for people who want to observe wildlife at 747405 (1). There is a golf course for golf enthusiasts at 668428 (1). There are various camp/caravan sites for people to stay whilst visiting the various attractions in the area (1).</p> |

| Question |  |  | General Marking Instructions for this type of question                                                                                                                                                                                                                                                      | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| 4.       |  |  | <p><b>No marks for description. Answers must explain the conflict.</b></p> <p>One mark for each valid point.<br/>Two marks for a developed point.<br/><b>Maximum 3 marks if no named area.</b></p> <p>If conflict between more than two land uses is explained, then award marks for the best two only.</p> | 4        | <p>Answers will vary depending upon the land uses chosen.</p> <p><b>Problems between tourists and farmers:</b><br/>In the Cairngorms, tourists can disrupt farming activities as walkers leave gates open, allowing animals to escape (1). Tourists' dogs can worry sheep if let off their lead (1). Stone walls are damaged by people climbing over them instead of using gates/stiles (1). Noisy tourists can disturb sheep especially during breeding season (1). Farmers may restrict walkers access at certain times eg lambing season (1). Farm vehicles can slow up tourist traffic on roads (1) and parked cars on narrow country roads can restrict the movement of large farm vehicles (1).</p> <p><b>Problems between industry and tourists:</b><br/>Tourists want to see the beautiful and unusual scenery of the Yorkshire Dales but quarries spoil the natural beauty of the landscape (1). Lorries used to remove the stone endanger wildlife and put visitors off returning to the area (1). This threatens local tourist-related jobs eg in local restaurants (1). The large lorries needed to remove the quarried stone cause air pollution which spoils the atmosphere for tourists (1). Lorries cause traffic congestion on narrow country roads which slows traffic and delays drivers (1). The peace and quiet for visitors is disturbed by the blasting of rock (1). Some wildlife habitats may also be disturbed by the removal of rock (1).<br/>Or any other valid point.</p> |



| Question |  |  | General Marking Instructions for this type of question                                                                                                                                                                                                                                                                                                                                       | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.       |  |  | <p>One mark for each valid point.<br/>Two marks for a developed point.</p> <p><b>No marks for description.</b></p> <p>Answers must refer to factors which affect <u>average</u> UK temperatures, not to factors which affect temperatures at one particular time such as the passage of a depression or a tropical continental air mass.</p> <p>Credit references to urban heat islands.</p> | 4        | <p>South-East England is usually warmer because it is closer to the Equator (1). This is due to intense heating from the sun (1) because sun rays are more concentrated (1). Places in Northern Scotland eg Wick, are colder because they are closer to the North Pole(1). This is due to a lack of insolation from the sun as the rays are less concentrated (1) and reflection of heat by the snow and ice (1). Places located on flat low-lying land are warmer eg Central Scotland, because temperatures increase as altitude decreases and places higher up ie mountainous regions are colder (1) because temperature decreases by 1°C for every one hundred metres in height (1). Places which are south facing are warmer because they get more sun (1) and places which are north facing are colder because they experience cold northerly winds (1). Western coastal areas are warmer because of a warm ocean current (1) (The North Atlantic Drift) and due to the prevailing South-Westerly winds that are warmed as they pass across this warm ocean current (1). In summer, places closer to the sea are cooler and in winter they are warmer because the sea heats up slowly in summer and cools slowly in winter (2).</p> <p>Or any other valid point.</p> |

## Section 2: Human Environments

| Question |     | General Marking Instructions for this type of question                                                                                                                                                                                                                                                                                                                      | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.       | (a) | <p>One mark per valid point.</p> <p>Do not accept grid-iron street pattern, lack of open space or hospital.</p> <p>No marks for grid references (grid reference is given in the question).</p> <p><b>No marks for features outwith the square (e.g. cathedral)</b></p>                                                                                                      | 3        | <p>Main roads lead into this square (1) there is a bus station (1) and two railway stations (1) tourist information centre (1) several churches (1) museum (1)</p> <p>Or any other valid point</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|          | (b) | <p>One mark per valid point. Two marks for a developed point.</p> <p>Maximum of one mark for an appropriate grid reference.</p> <p>If no reference to the Ordnance Survey map then mark out of three.</p> <p>Do not credit the same point twice (e.g. the land is flat and suitable for building).</p> <p><b>Answers must be explanatory. No marks for description.</b></p> | 5        | <p>The land is flat so easy to build on (1) there is space available for expansion (1) eg expansion of the motor works at 163823 (1). There are good transports links like the M42 allowing people and products access to and from the area (1). A rail link with Birmingham International Rail Station gives easy access to the airport (1). There are many road junctions and intersections connecting the area to other areas and less traffic congestion as it is away from Birmingham city centre (2). The land is on the edge of Birmingham so will be cheaper encouraging housing estates like Sheldon to be built (1). The cheaper land allows the houses to be bigger with cul-de-sacs, gardens etc.(1) the houses can provide a source of labour for the airport, motor works and the business park (1)</p> <p>Or any other valid point</p> |

| Question |  |  | General Marking Instructions for this type of question                                                                                                                                                                                                                                                                         | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| 7.       |  |  | One mark per valid point. Two marks for a developed point.                                                                                                                                                                                                                                                                     | 6        | <p>Contraception and family planning is widely available (1). Later marriages are more common which results in fewer children (1). People no longer choose to have lots of children as improved medical care and advances in medicine (1) has resulted in most children surviving at birth (1). Developed countries have the money to invest in medical care which reduces the infant mortality rate thus causing the birth rate to fall (1). Children are expensive so the greater number of children the bigger the financial burden (1). Women want careers so put off having children to a later age (1) or limit the size of their families to give them a reasonable standard of living (1). Sex education in schools helps to lower birth rates (1).</p> <p>Or any other valid point.</p> |
| 8.       |  |  | <p>One mark per valid point. Two marks for a developed point.</p> <p><b>Mark out of five if no reference to a specific developing world city.</b></p> <p>Candidates may refer to more than one city.</p> <p>Max. of 1 mark for a list of improved facilities (e.g. better toilets, better housing, better water supplies).</p> | 6        | <p>For example in Rocinha (Rio), the former wooden shacks have been upgraded to permanent dwellings with some modern services (1). Residents constantly improve their homes through a process of 'self-help' (1) where the residents are provided with materials like bricks (1). Some prefabricated houses have been built by the Brazilian government (1) with basic facilities like toilets, electricity and running water (1). The residents have been given the legal rights to the land (1), roads have been built into/ or improved in the favela (1) allowing services like rubbish collections to take place (1), there are now a few health clinics and schools provided (1).</p> <p>Or any other valid point.</p>                                                                     |

### Section 3: Global Issues

| Question |     | General Marking Instructions for this type of question                                                                                                                                                                                                                                            | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| 9.       | (a) | <p>One mark for each valid point. Two marks for a developed point.</p> <p><b>Maximum of two marks if no reference to figures</b></p>                                                                                                                                                              | 4        | <p>The overall trend is that the amount of Arctic Sea ice has decreased between 1979 and 2013 (1) from (around) 7 million square kilometres to (about) 5 million square kilometres (1). There has been a fluctuation in the extent of sea ice in certain years (1) eg in 2013, the amount of sea ice increased from 3.75 million square kilometres in 2012 to 5 million square kilometres (1) whereas, between 2006 and 2007 there was a sharp decrease (1) from 6 million square kilometres to 4.25 million square kilometres (1). Or any other valid point.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|          | (b) | <p>One mark for each valid point. Two marks for a developed point. Marks for explanatory points only not for description</p> <p>Credit can still be given for effects linked to melting sea ice.</p> <p>Answers must refer to effects not causes.</p> <p>Effects can be positive or negative.</p> | 6        | <p>Increased temperatures are causing ice caps to melt so Polar habitats are beginning to disappear (1). Melting ice causes sea levels to rise (1) threatening coastal settlements (1). An increase in sea temperatures causes the water to expand, compounding the problem of flooding (1). Global warming could also affect weather patterns, leading to more droughts (1) crop failures and problems with food supply (1); flooding, causing the extinction of species (1) and more extreme weather, eg tropical storms (1). Tourism problems will increase as there will be less snow in some mountain resorts (1). Global warming could threaten the development of developing countries as restrictions on fossil fuel use may be imposed to slow the rate of increasing CO<sub>2</sub> levels (1). In the UK, tropical diseases like malaria may spread as temperatures rise (1). Plants growth will be affected and some species will thrive in previously unsuitable areas (1). Higher temperatures may cause water shortages (1). Or any other valid point.</p> |

| Question |     | General Marking Instructions for this type of question                                                                                                                                                                                                                                                                                                                                                                              | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.      | (a) | <p><u>Trends/changes</u> in deforestation to be described with reference to figures and years.</p> <p>One mark for each valid point.<br/>Two marks for a developed point.<br/><b>Maximum of two marks if no reference to figures</b></p>                                                                                                                                                                                            | 4        | <p>Overall the amount of deforestation in Peru 2004-2012 has decreased (1) from just under 3 million ha to 750,000 ha (1). The deforestation rate declined rapidly from 2004 to 2007 (1). Deforestation increased from 2007 to 2008 peaking in Peru at 1 500 000 hectares per year (1). Again Peru experienced a decline in deforestation rates from 2008 to 2009 by over 500,000 ha (1). From 2009 to 2010 deforestation rates rose to around 1 400 000 ha(1), before declining to around 750 000 hectares per year in 2012 (1).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|          | (b) | <p>Reference should be made to a named area studied.<br/>Answer will depend on area referred to.</p> <p>One mark for each valid point.</p> <p><b>No marks for description</b></p> <p>Two marks for a developed point.</p> <p>Mark out of five if no reference to specific area</p> <p>Candidates can gain full marks by referring to only environment or people.</p> <p>Marks can be awarded for positive and negative impacts.</p> | 6        | <p>New industries have led to the expansion of towns such as Anchorage in Alaska which have grown to accommodate workers (1). Although these industries provide employment (1), these developments spoil the appearance of the natural landscape (1). New roads have been built to transport people and goods. This increases the number of vehicles in the tundra creating noise and air pollution (1). But also improves access to locals (1).</p> <p>Oil is a very important industry in Alaska. The building of oil platforms and oil pipelines has resulted in damage to tundra vegetation and wildlife (1). In some areas, the Trans-Alaskan oil pipeline has been built on natural migration or hunting routes for animals, which hinders the natural movement of caribou (1). Local Inuit people have also had their way of life disrupted as they must detour around the pipeline (1) and may no longer have access to their traditional hunting grounds (1). Local people were promised jobs in the industry, but few jobs are available for locals (1).</p> <p>Burst pipes have spilt hundreds of thousands of gallons of crude oil in Alaska, devastating this fragile environment (1). Oil spills have also been responsible for pollution in the region (1), such as the Exxon Valdez disaster (1).</p> <p>Any damage to the tundra landscape is slow to recover, as the short growing season means that bulldozer tracks from the oil and natural gas industries could take centuries to restore (1).</p> <p>Pollution from mining and oil drilling has contaminated the air, lakes and rivers (1).</p> <p>Any other valid point.</p> |

| Question |     | General Marking Instructions for this type of question                                                                                                                                                                                                                                                                                        | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 11.      | (a) | <p>One mark per valid point.</p> <p><b>No marks for explanation .</b></p> <p><b>Max 1 mark for a list of places.</b></p> <p>Do not accept reference to areas (or cities) which are <b>not</b> affected by earthquakes.</p>                                                                                                                    | 4        | <p>Most cities are located on or near plate boundaries (1) where seismic activity is highest (1). Most earthquake threatened cities are found in developing countries (1) like Indonesia (1). A large number of threatened cities are found in China (1). Three cities in Africa are at risk (1). All threatened cities in the USA are found on the west coast (1) with a cluster around San Francisco/Los Angeles (1).</p> <p>Or any other valid point.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|          | (b) | <p>One mark per valid point. Two marks for a developed point.</p> <p><b>No marks for description</b></p> <p>Mark out of five if no reference to specific area.</p> <p>Marks may be awarded for reference to warnings given/ evacuations taking place prior to tsunamis.</p> <p>A good answer which refers only to aid may get full marks.</p> | 6        | <p>In Japan people take part in earthquake drills to practise what to do in the event of an earthquake (1) giving them a better chance of survival (1). The government warn people, using text messages and TV, giving them the chance to move to a safer place (1). Earthquake resistant buildings reduce the number of people trapped or killed (1). Sprinkler systems and gas cut off valves prevent fires spreading reducing the number of people injured and buildings destroyed (1). People living in earthquake prone areas have emergency plans in place and emergency supplies such as bottled water and tinned food are stockpiled to ensure they have vital supplies to survive in the event of an earthquake (2). In the event of an earthquake short term aid in the form of food, medicine and shelter is sent to the area to treat the injured (1).</p> <p>Or any other valid point.</p> |

| Question |     | General Marking Instructions for this type of question                                                                                                                | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12.      | (a) | <p>One mark per valid point. Two marks for a developed point</p> <p><b>No marks for explanation</b></p> <p><b>Maximum of two marks if no reference to figures</b></p> | 4        | <p>The value of exports from developed world countries to developing world countries is \$738bn (1) whereas there is only \$650bn worth of goods exported from developing to developed world countries (1). That is a difference of \$88 billion (1). The value of trade between developing world countries is \$383bn (1). The value of trade between developed world countries is \$2251bn (1). There is more trade between developed world countries than between developing countries (1); it is \$1868bn more (1).</p> <p>Or any other valid point.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|          | (b) | <p>One mark per valid point. Two marks for a developed point.</p> <p>If no reference to an area of study (e.g. Africa/EU), then mark out of 5.</p>                    | 6        | <p>There is a big imbalance in the pattern of trade between the developing and developed world; this can reinforce differences in wealth between areas such as the EU and Africa (1); African countries export mainly primary products such as oil or cocoa beans for comparatively low prices but import mainly processed goods such as vehicles for much higher prices (1) which can result in a trade deficit for them (1); this can increase levels of poverty within African countries and cause difficulties for the economy as well (2); often the producers such as cocoa farmers in Africa receive very low wages and so struggle to maintain a decent standard of living (2); wealthy European countries profit from selling expensive manufactured goods to African countries (1), helping to keep a much higher standard of living for their citizens (1); often, exploitation of primary products in African countries can lead to serious environmental damage, such as logging which has caused deforestation (1), resulting in the loss of areas of rainforest as well as the destruction of animal habitats (1).</p> <p>Or any other valid point.</p> |

| Question |     | General Marking Instructions for this type of question                                                                                                                                                                                                                                                                                               | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 13.      | (a) | <p>One mark for each valid point.</p> <p>Two marks for a developed point.</p> <p><b>No marks for explanation</b></p> <p><b>Maximum of two marks if no reference to figures.</b></p>                                                                                                                                                                  | 4        | <p>There has been a fairly steady increase in visitor numbers since 1995 (1) from around 525 million reaching 1 billion in 2013 (1). There were only 2 years where the numbers decreased slightly ie in 2003 (1) when it dropped to just under 700 million (1) and in 2009, dropped to under 900 million (1). The period with the largest increase was the 5 years between 1995 - 2000 (1) whereas the slowest increase has been in recent years from 2010 (1)</p> <p>Or any other valid point.</p>                                                                                                                                                                                                                                                                                                                                                   |
|          | (b) | <p>Reference should be made to a named tourist area. Mark out of 5 if no reference made.</p> <p>Answers will depend on area referred to, which could include places in the UK or Europe.</p> <p>One mark for each valid point.</p> <p>Two marks for a developed point.</p> <p>Mark out of five if both people and environment are not mentioned.</p> | 6        | <p>If Costa Rica cloud forest chosen:</p> <p>Eco-tourism raises local as well as international awareness of natural environment (1) such as wildlife and vegetation (1). Developing countries now want to conserve their fragile environments and view eco-tourism as a significant means of generating income (1). Developed countries want to help developing countries conserve their fragile environments by promoting sustainable/eco-tourism (1). Tourists are now more environmentally conscious and want to help protect fragile environments for future generations (1). Eco-tourism provides work and opportunities for local people (1) hence improving their standard of living (1) encourages local enterprise and improvement schemes (1) promoting awareness of local culture and traditions (1).</p> <p>Or any other valid point.</p> |



| Question |     | General Marking Instructions for this type of question                                                                                                                                                                                                  | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| 14.      | (a) | <p>One mark per valid point and two for a developed point.</p> <p><b>Maximum of <u>three</u> marks if no reference to figures</b></p>                                                                                                                   | 4        | <p>Male deaths from heart disease are most common in Eastern Europe (1). Russia for example, has a rate of 444-841 per 100,000 (1). This compares to only 120-238 in the UK (1). Canada, the USA and Mexico have some of the lowest rates (1), with under 120-238 per 100,000 (1). Many central African countries have rates of 363-443. (1)</p> <p>Or any other valid point.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|          | (b) | <p><b>No marks for simple description of any control methods.</b> Answers must be explanatory.</p> <p>One mark per valid point and two for a developed point.</p> <p>If more than one disease mentioned, give marks for the highest scoring answer.</p> | 6        | <p>If <b>pneumonia</b> chosen:</p> <p>Antibiotics are used to treat any bacterial lung infections (1) and patients are encouraged to drink plenty in order to avoid dehydration (1); in severe cases a drip may be required to restore the right level of salts and fluids quickly (1); paracetamol is used to ease the effects of fever and/or headaches (1); introducing more community-based health workers helps to control the incidence of pneumonia as children with the disease are more likely to be diagnosed and treated quickly (1); this can often help to save lives (1). Vaccinations are being increasingly used in developing world countries to protect children against common infections such as flu (1); adequate nutrition helps to increase a child's natural defences against disease and so education about this also helps to reduce pneumonia (1).</p> <p>Or any other valid point.</p> |

| Question |     | General Marking Instructions for this type of question                                                                                                 | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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|          | (b) | (cont)                                                                                                                                                 |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|          |     | <p>No marks awarded for effectiveness of control methods.</p> <p><b>Max one mark for reference to named examples (of drugs/ insecticides etc).</b></p> |          | <p>If <b>kwashiorkor</b> is chosen:</p> <p>The main method of managing kwashiorkor is education about the need for a well-balanced diet, so that children don't develop the disease in the first place (1); by educating communities they can be encouraged to grow different food types to increase protein intake (1); this might include crops such as cashews, peanuts, lentils or sunflower (1) and might also involve advice about constructing irrigation schemes to help crops grow better in times of drought (1); education about family planning also helps to reduce the number of children per family, making more food available per child (1).</p> <p>For children who have kwashiorkor it is important to give vitamin and mineral supplements as salt and mineral levels in their blood stream may be dangerously low (1); Zinc supplements might also be administered to help the skin recover (1). Small amounts of food are reintroduced slowly, such as carbohydrates to give energy (1) and protein rich foods to help the child's body recover (1).</p> <p>Or any other valid point.</p> |

| Question |     |  | General Marking Instructions for this type of question | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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|          | (b) |  | (cont)                                                 |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|          |     |  |                                                        |          | <p>If <b>malaria</b> chosen:</p> <p>Anti-malarial drugs kill blood parasites (1) Chloroquine is an example of this.(1) Insecticides, such as malathion destroy the female anopheles mosquito (1).</p> <p>Draining all breeding areas eradicates larvae (1), planting eucalyptus trees to soak up moisture removes breeding ground (1). Water can also be released from dams to drown immature larvae (1). Mustard seeds can be used to drag larvae below the surface to drown them (1). Small fish can be introduced to eat larvae and provide a cheap protein source (1). Genetic engineering of sterile male mosquitoes reduces mosquitoes (1).</p> <p>Health education teaches people about how to protect themselves from being bitten (1). Preventative bed nets are cheap and effective at stopping mosquitos biting at night (1). New treatments have also been developed which seem to be more effective such as artemesinin/ACT because malaria parasite is not yet resistant to them (1) .</p> <p>Or any other valid point.</p> |

| Question |     |  | General Marking Instructions for this type of question | Max Mark | Specific Marking Instructions for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|          | (b) |  | (cont)                                                 |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|          |     |  |                                                        |          | <p>If <b>cholera</b> chosen:</p> <p>One of the main ways to reduce or control the spread of cholera is to improve sanitation which stops disease from spreading (1). Providing wells and pipes makes drinking water safe and clean (1). Health Education encourages people to wash hands often with soap and safe water preventing infection as does building and use of latrines (2). Because of contaminated water people should cook their food well and eat it hot (1). Food stuffs should be kept covered and fruit and vegetables should be peeled to prevent contamination (2). Cholera is an easily treatable disease.</p> <p>The main ways to treat cholera are either a simple drink made from 1 litre of safe water, 6-8 teaspoons of sugar and 1/2 teaspoon of salt helps to rehydrate sufferers so that they can fight off the disease (2) or re-hydration tablets, if available (1). In especially severe cases, intravenous administration of fluids may be required to save the patient's life (1). Treatment with antibiotics is recommended for severely ill patients to help fight the infection (1).</p> <p>Or any other valid point.</p> |

[END OF MARKING INSTRUCTIONS]